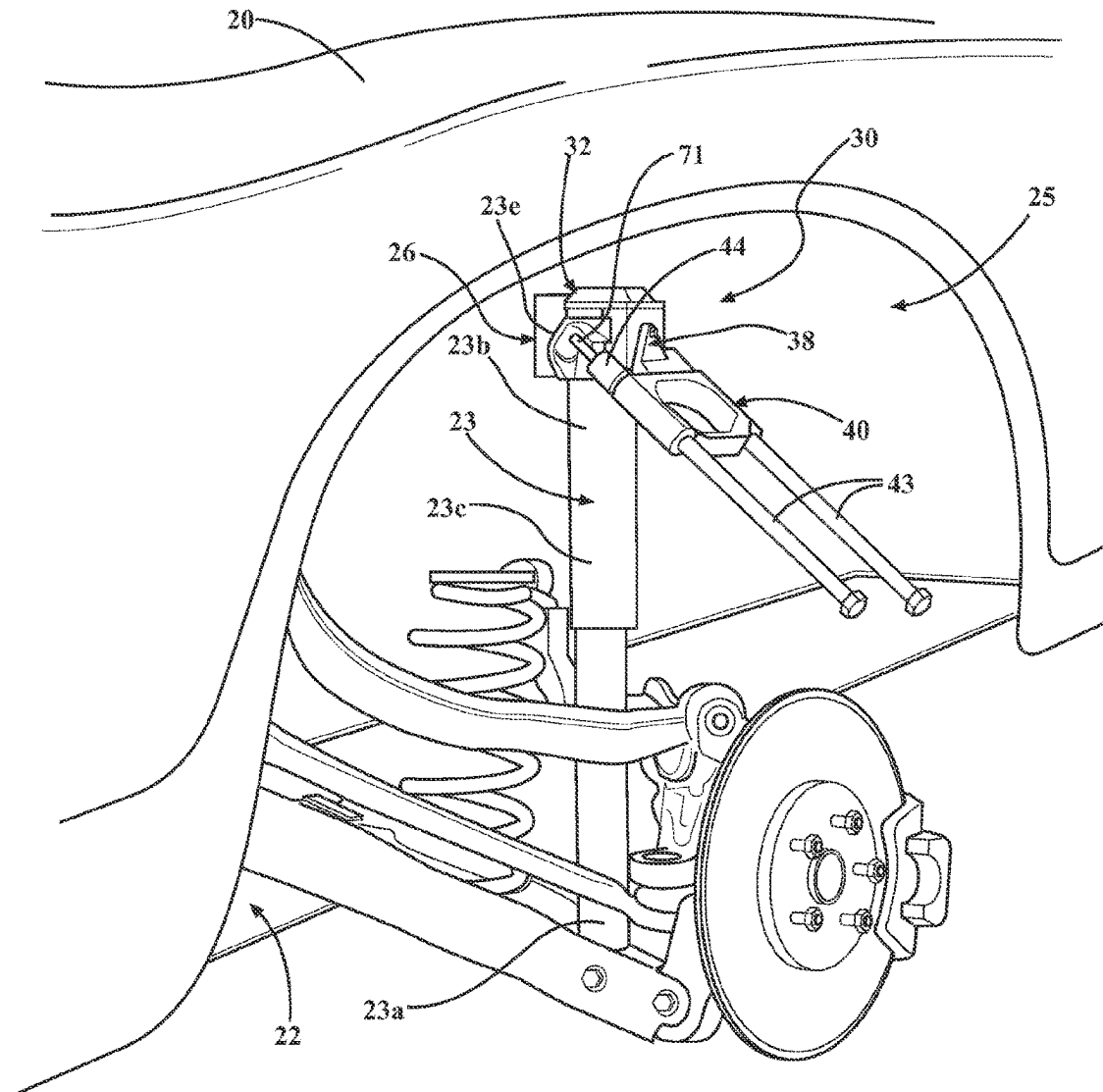




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Reshad et al.(10) **Pub. No.: US 2025/0282433 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SHOCK ABSORBER ATTACHMENT JIG****Publication Classification**(71) Applicant: **Toyota Motor Engineering & Manufacturing North America, Inc.**,
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CPC **B62D 65/026** (2013.01)(72) Inventors: **Samuel Garrigus Reshad**, Newburgh,
IN (US); **Chase O. Muncy**, Haubstadt,
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IN (US)(57) **ABSTRACT**

A shock absorber attachment jig includes a cap including a body portion and a flap extending from the body portion, and a fastener-receiving sub-assembly structured to be attached to the flap. The flap is structured to be rotatable with respect to the body portion so as to enable rotation of the sub-assembly with respect to the cap body portion when the sub-assembly is attached to the flap. When attached to a shock absorber, the jig enables movement of a fastener received in the sub-assembly into an assembly orientation of the fastener, thereby facilitating application of the fastener to the shock absorber and a vehicle body to attach the shock absorber to the body.

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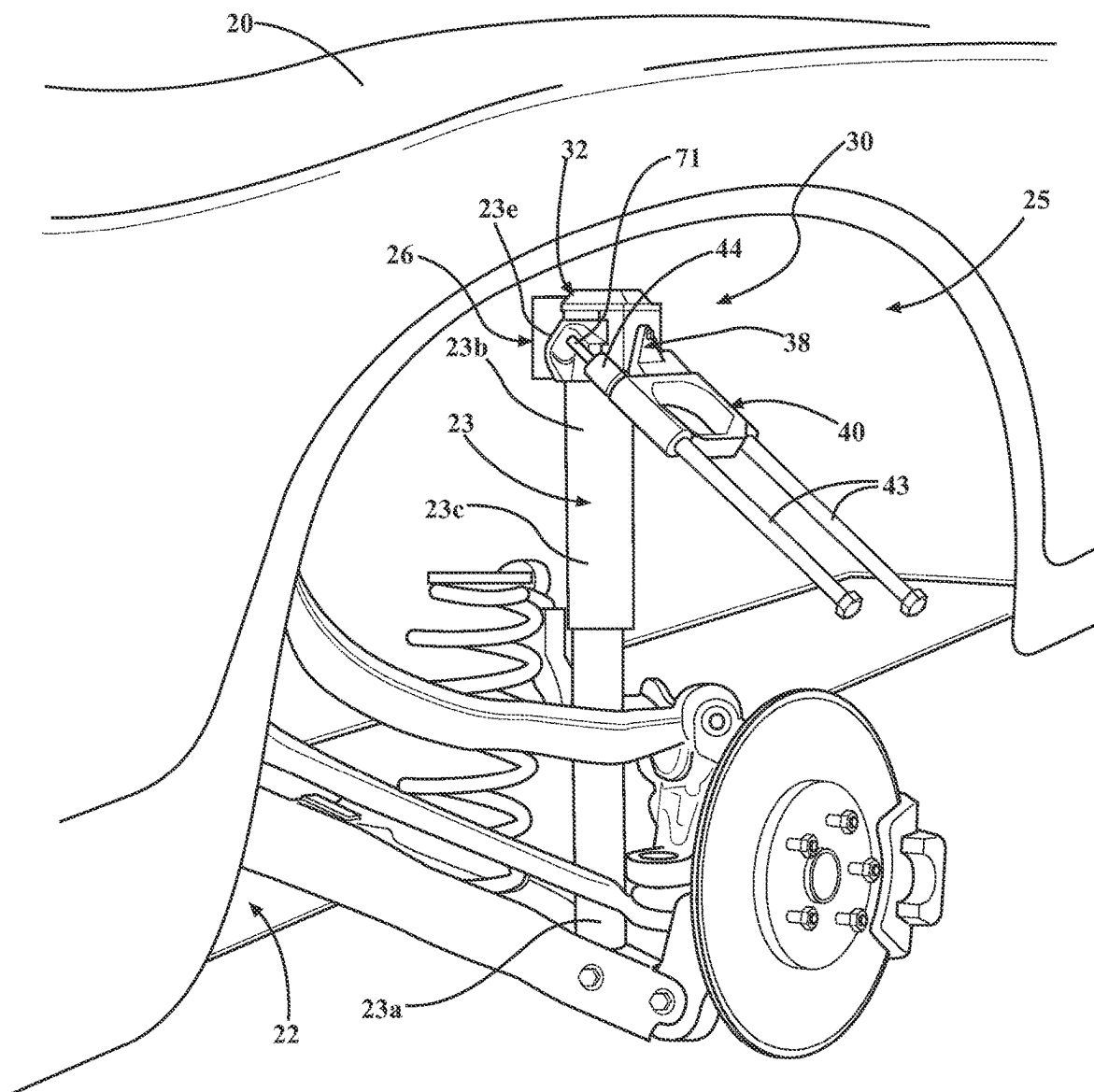


FIG. 1

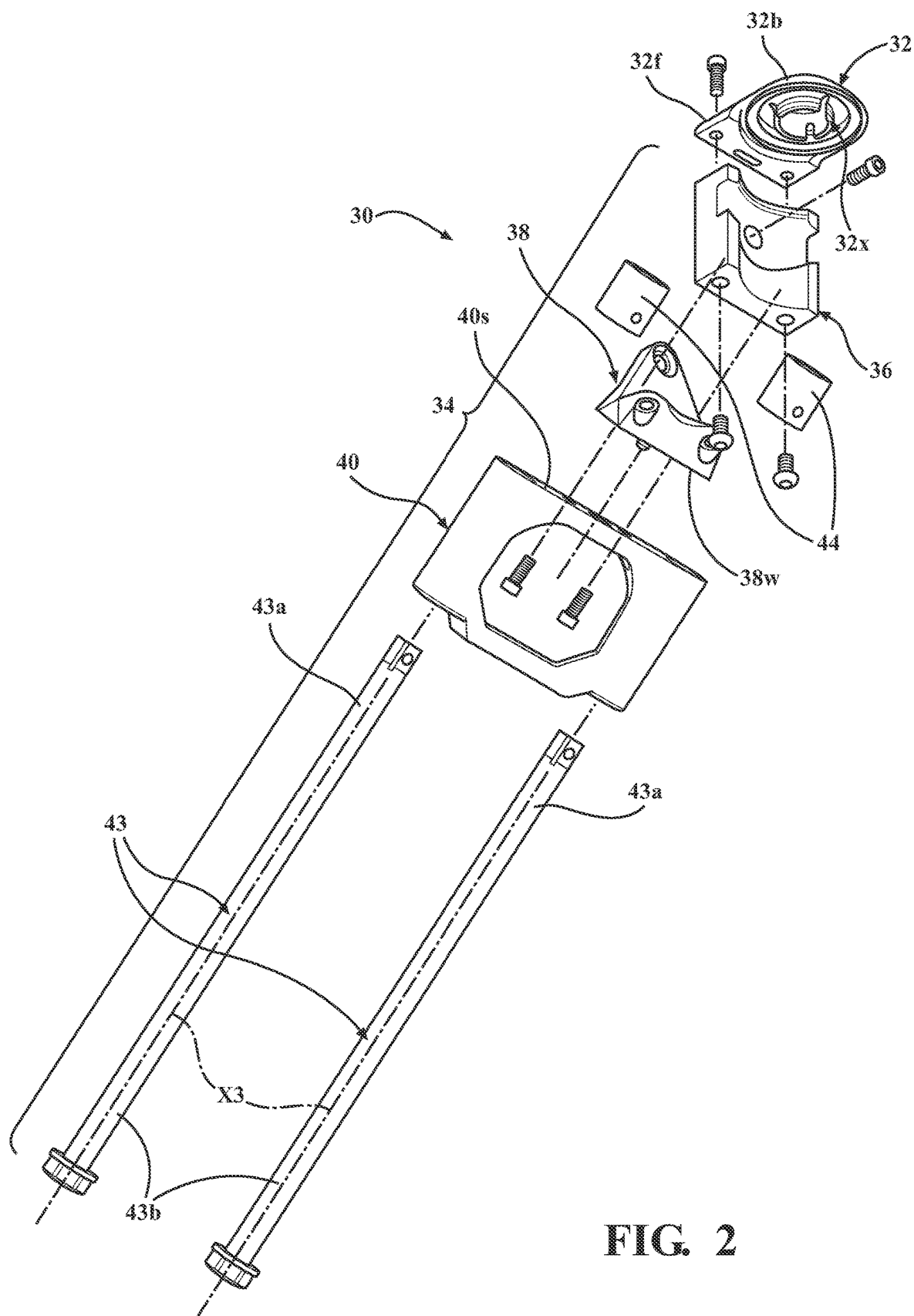


FIG. 2

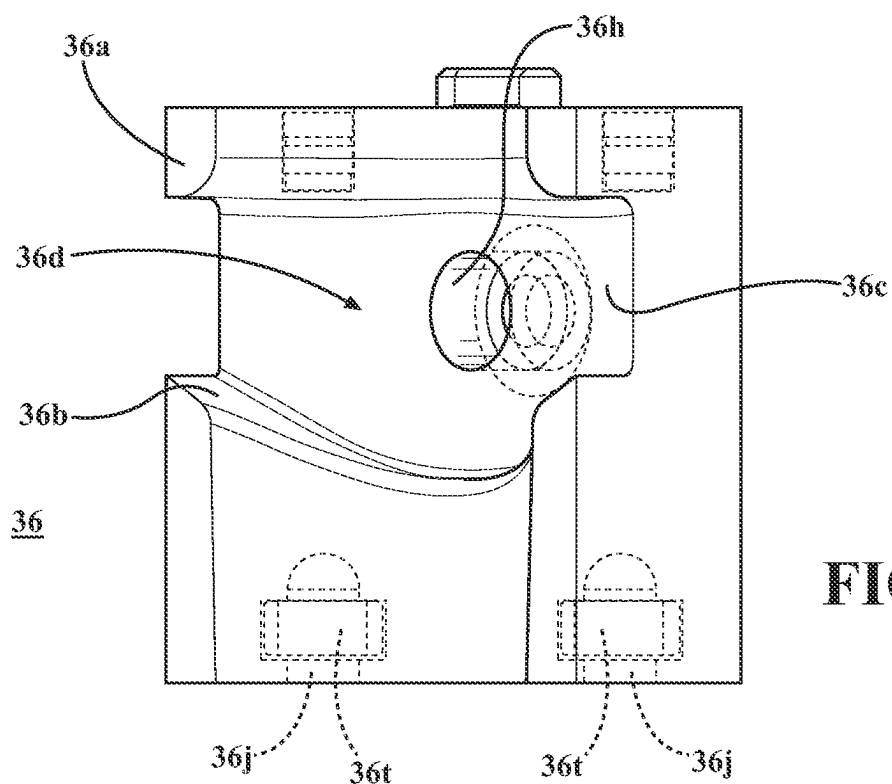


FIG. 3

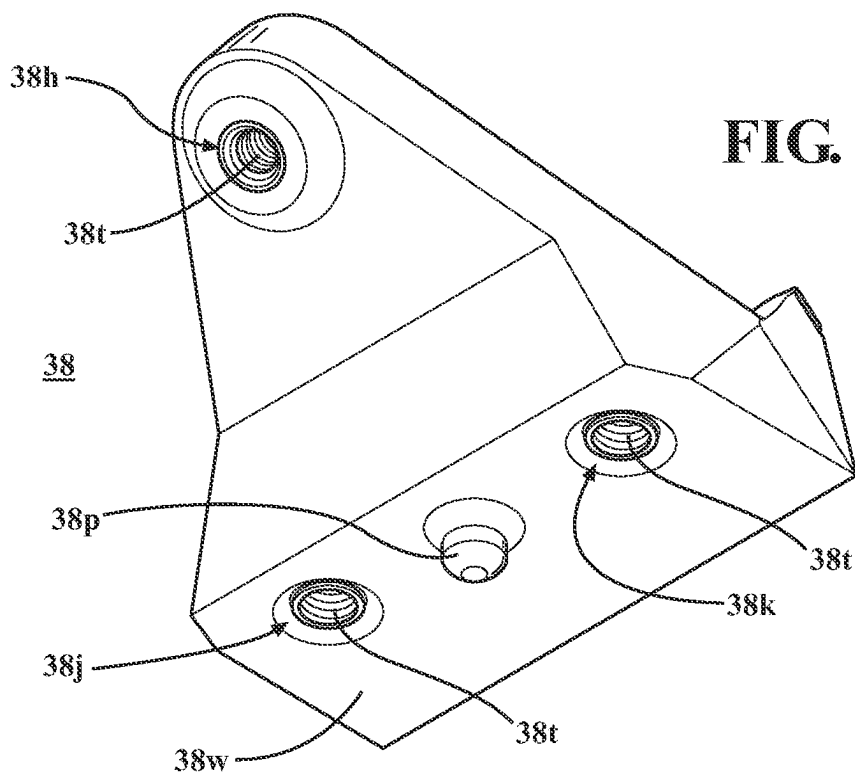
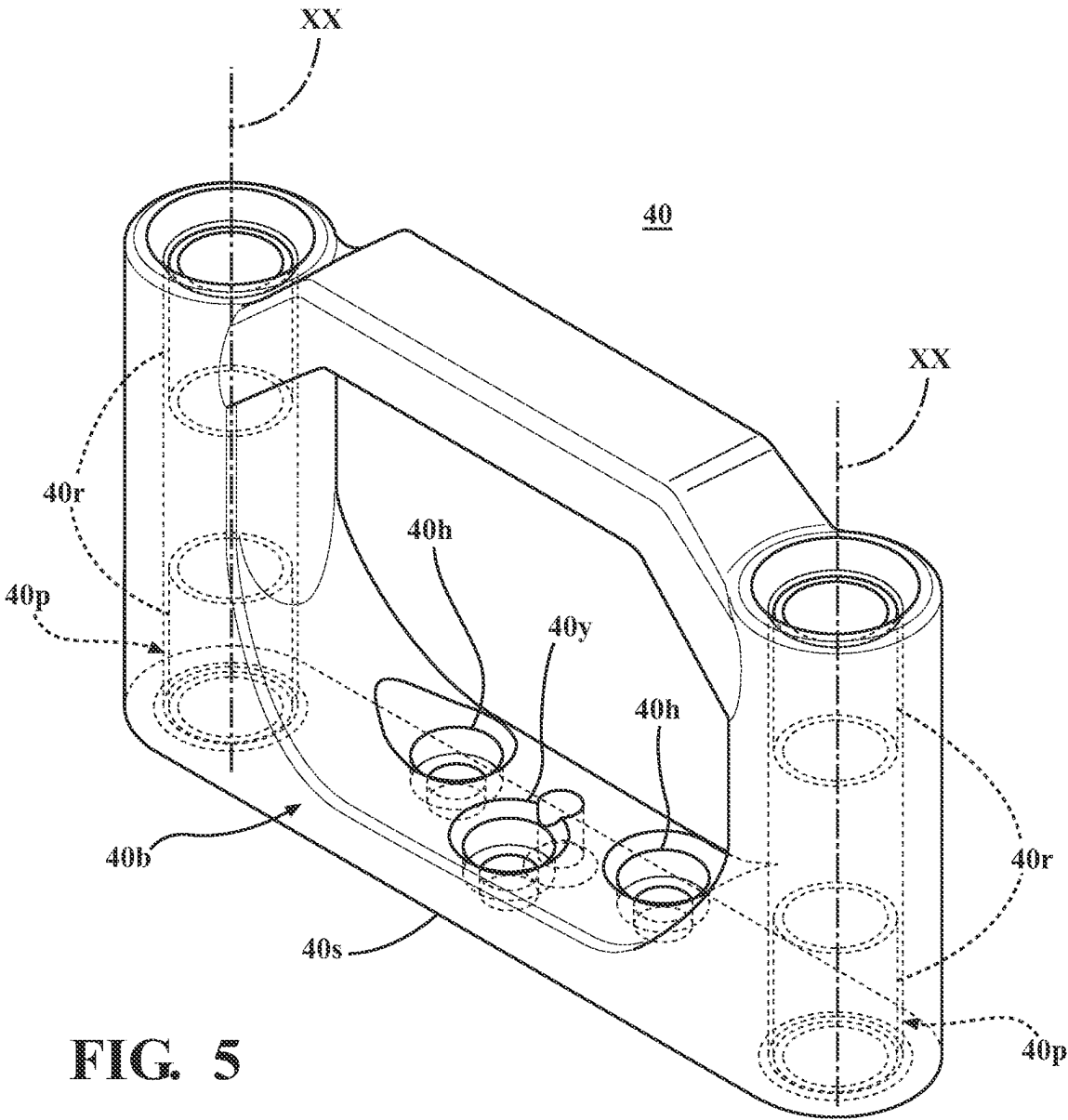
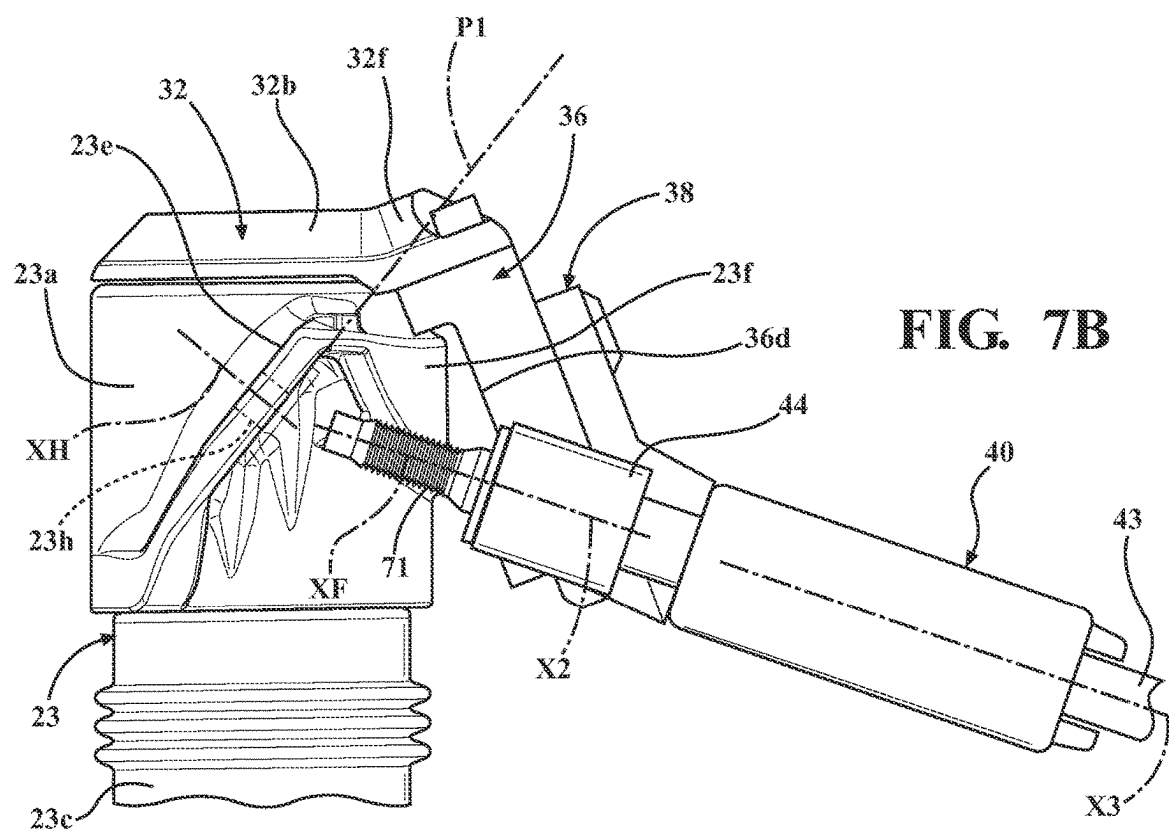
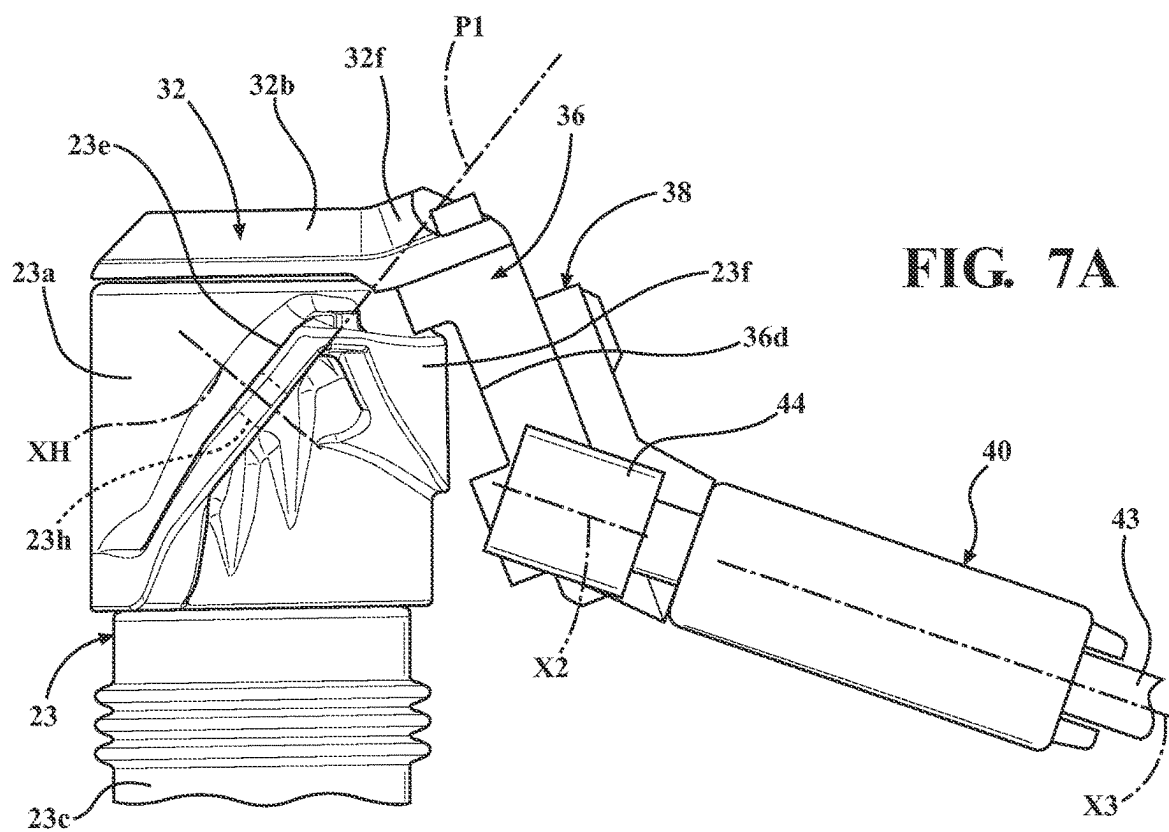
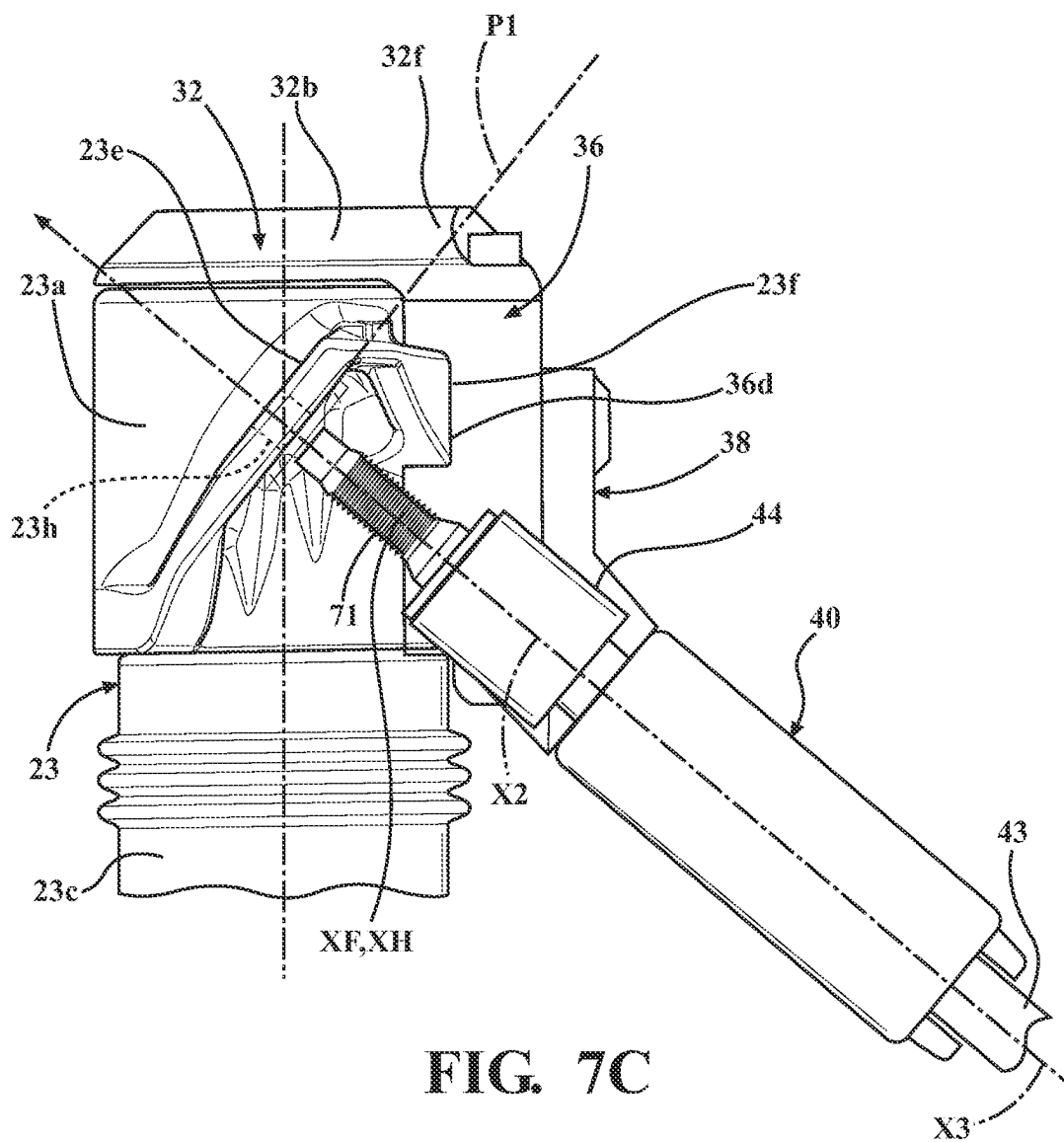


FIG. 4







SHOCK ABSORBER ATTACHMENT JIG

TECHNICAL FIELD

[0001] The present invention relates to jigs used for assembly of vehicles and, more particularly, to a jig used for attaching a portion of a shock absorber to a vehicle body during vehicle assembly.

BACKGROUND

[0002] An ongoing problem in assembly of motor vehicles is limited physical access of assembly personnel to confined or restricted spaces between portions of the vehicle. It is often time-consuming and physically very difficult to properly position, orient, and apply mechanical fasteners where and when they are needed to attach vehicle components to each other. It may also be difficult for a user to actually see into the area where the fasteners are to be applied. In particular, due to the structure of a vehicle body in the area where a shock absorber is to be attached to the body, and the need to attach other components to the vehicle frame and body prior to attaching the shock absorber to the vehicle, it may be very difficult to attach a shock absorber to the vehicle body at a designated attachment location during vehicle assembly.

SUMMARY

[0003] In one aspect of the embodiments described herein, a shock absorber attachment jig is provided for attaching a portion of a shock absorber to a vehicle. The jig includes a cap including a body portion and a flap extending from the body portion, and a fastener-receiving sub-assembly structured to be attached to the flap. The flap is structured to be rotatable with respect to the body portion so as to enable rotation of the sub-assembly with respect to the cap body portion when the sub-assembly is attached to the flap.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale. Also, unless otherwise stated or shown, the same or similar elements shown in different views may be given the same or similar reference numerals or designations.

[0005] FIG. 1 is a schematic perspective view of a shock absorber attachment jig in accordance with an embodiment described herein, shown attached to a vehicle during assembly of the vehicle.

[0006] FIG. 2 is a schematic exploded view of the shock absorber attachment jig shown in FIG. 1.

[0007] FIG. 3 is a schematic side perspective view of a bracing portion of the shock absorber attachment jig shown in FIG. 1.

[0008] FIG. 4 is a schematic bottom perspective view of a connecting element of the shock absorber attachment jig shown in FIG. 1.

[0009] FIG. 5 is a schematic perspective view of an extender guide of the shock absorber attachment jig shown in FIG. 1.

[0010] FIG. 6 is a schematic perspective view of a portion of a fastener-receiving sub-assembly of the shock absorber attachment jig shown in FIG. 1.

[0011] FIGS. 7A-7C are schematic side views showing attachment of the shock absorber attachment jig to a shock absorber and rotation of the fastener-receiving sub-assembly to move a fastener into an assembly orientation of the fastener for application of the fastener to the shock absorber and a shock tower of a vehicle.

DETAILED DESCRIPTION

[0012] A shock absorber attachment jig includes a cap including a body portion and a flap extending from the body portion, and a fastener-receiving sub-assembly structured to be attached to the flap. The flap is structured to be rotatable with respect to the body portion so as to enable rotation of the sub-assembly with respect to the cap body portion when the sub-assembly is attached to the flap. When attached to a shock absorber, the jig enables movement of a fastener received in the sub-assembly into an assembly orientation of the fastener, thereby facilitating application of the fastener to the shock absorber and a vehicle body to attach the shock absorber to the body.

[0013] The various elements set forth in the following description may be fabricated from any material or materials (e.g., metals, polymers, etc.) suitable for the purposes described herein. Also, if desired, suitable threaded metallic inserts may be secured in any of the components described herein to facilitate the use of threaded fasteners attaching and securing the components to each other.

[0014] FIG. 1 is a schematic perspective view of a shock absorber attachment jig 30 in accordance with an embodiment described herein, shown attached to a vehicle 20 during assembly of the vehicle. Vehicle 20 may include a frame 22 and a body 25 configured for mounting on the frame 22 during vehicle assembly.

[0015] Referring to FIG. 1, as known in the pertinent art, a shock tower 26 may be attached to the vehicle body 25 to provide an attachment point for an upper end of a suspension damper, such as a shock absorber 23. The shock tower 26 may include one or more threaded holes into which fastener(s) may be inserted to secure attachment ears 23e of the shock absorber 23 to the shock tower 26, thereby mounting the shock absorber 23 to the vehicle body 25.

[0016] The shock absorber 23 may have a first end 23a, a second end 23b opposite the first end 23a, and a longitudinal body 23c extending between the first and second ends. The first end 23a may be a lower end of the shock absorber 23 when the shock absorber 23 is coupled to an associated axle or other portion of the vehicle mounted to the frame 22. The second end 23b may be an upper end of the shock absorber 23 when the second end 23b is connected to the shock tower 26.

[0017] Referring to FIGS. 1 and 7A-7C, one or more attachment ears 23e may extend outwardly from the shock absorber body 23c. Each attachment ear 23e may have a through-hole 23h extending therethrough to enable attachment of the attachment ear 23e to the shock tower 26 using mechanical fasteners as described herein. Where a pair of ears 23e are used as shown in FIGS. 1 and 7A-7C, the ears may extend from opposite sides of the shock absorber body

23c so that surfaces of the ears **23e** including openings into the through-holes **23h** in a direction DI toward the vehicle body **25** are coplanar (within applicable tolerance limits) along a plane P1. A flange **23f** may also extend along an exterior of the shock absorber body **23c** between the attachment ears **23e**.

[0018] FIG. 2 is a schematic exploded view of the shock absorber attachment jig **30** shown in FIG. 1. Referring to FIGS. 1 and 2, the jig **30** may include a cap **32** structured for attachment to the shock absorber **23** at or near an upper end **23b** of the shock absorber. Cap **32** may include a body portion **32b** and a flap **32f** extending from (and formed integrally with) the body portion **32b**. In particular arrangements, as shown in FIGS. 1 and 6, the cap body portion **32b** may incorporate attachment features (generally designated **32x**) structured to form an interference fit with features (not shown) already present on the upper end **23b** of the shock absorber **23**, thereby securing the cap **32** to the shock absorber **23** until the shock absorber is attached to the vehicle body **25** using mechanical fasteners as described herein.

[0019] The flap **32f** may incorporate tabs, holes, and/or other features enabling additional elements of the shock absorber attachment jig to be attached to the cap **32** as described herein. The cap **32** may be formed from a relatively flexible material so as to enable bending of the cap flap **32f** with respect to the cap body portion **32b** as described herein. Materials suitable for the cap **32** may have characteristics of both plastic and rubber. Examples of such materials include melt-processable thermoplastic elastomers such as various durometers of thermoplastic polyurethane (TPU). The cap **32** may be formed from any material suitable for the purposes described herein.

[0020] Referring to the drawings, the jig **30** may also include a fastener-receiving sub-assembly (generally designated **34**) structured to be attachable to the cap flap **32f**. The fastener-receiving sub-assembly **34** may be structured to hold to mechanical fasteners **71** that will be used to attach the shock absorber **23** to the vehicle body **25**, and to maintain each fastener **71** in an assembly orientation of the fastener as described herein, when the jig **30** is mounted on the shock absorber **23** by attaching the cap **32** to the shock absorber.

[0021] Referring to FIGS. 1 and 3, the fastener-receiving sub-assembly **34** may include a bracing portion **36** structured to be attachable to the cap flap **32f**. The bracing portion **36** may include a first shoulder **36a**, a second shoulder **36b** positioned opposite the first shoulder **36a**, and a connecting wall **36c** extending between and connecting the first and second shoulders **36a**, **36b**. In combination, the first shoulder **36a**, the second shoulder **36b** and the connecting wall **36c** may define a cavity **36d** structured to receive the shock absorber flange **23f** therein when the jig **30** is attached to the shock absorber **23** as described herein.

[0022] In particular arrangements, the cavity **36d** may be structured so that the first and second shoulders **36a**, **36b** form a slight interference fit with portions of the flange **23f** when the flange is received in the cavity **36d** and the second shoulder **36b** is in physical contact with the shock absorber body **23c** adjacent the flange **23f**. This may help retain the first and second shoulders **36a**, **36b** in contact with the shock absorber body **23c** during movement of the shock absorber **23**, when the cap **32** is mounted on the shock absorber **23**.

[0023] Holes **36h** (in the connecting wall **36c**) and **36j** (in the second shoulder **36b**) may be provided to enable attachment of a connecting element **38** (described below) to the bracing portion **36**, using threaded fasteners for example. Suitable metallic threaded inserts **36i** may be pressed in, molded in, or otherwise secured in the holes **36h** and **36j** to facilitate use of threaded fasteners for connection.

[0024] The bracing portion **36** may be formed from a relatively rigid material, such as a suitable polymer, for example. The bracing portion **36** may be formed using any suitable method, such as *molding or 3-D printing, for example*.

[0025] Referring to FIGS. 1 and 4, the fastener-receiving sub-assembly **34** may include a connecting element **38** structured to connect the bracing portion and an extender guide **40** (described below). The connecting element **38** may be structured to be attachable to the bracing portion **36** by a fastener applied in a first connecting element hole **38h** and bracing portion hole **36h**. A fastener is “applied” when it is inserted into suitable hole(s) in one or more components to be connected or attached to each other by the fastener, and/or when the fastener is rotated or otherwise manipulated so as to secure the components together. A suitable metallic threaded insert **38i** may be pressed, molded, or otherwise secured in the connecting element hole **38h** to facilitate use of a threaded fastener for connection. Similarly, threaded inserts **38i** may be pressed, molded, or otherwise secured in second and third connecting portion holes **38j**, **38k** to facilitate use of a threaded fastener for the extender guide **40** to the connecting element **38**.

[0026] Connecting element **38** may include a bearing surface **38w** structured to abut and support the extender guide **40** in a predetermined orientation with respect to the bracing portion **36** when the extender guide **40** is attached to the connecting element **38** and the cap **32** is attached to the shock absorber **23** as described herein. A locating pin **38p** may extend from the bearing surface **38w** to aid in positioning the extender guide **40** with respect to the connecting element **38** for purposes of assembling the fastener-receiving sub-assembly **34**. The connecting element **38** may be formed from a relatively rigid material, such as a suitable polymer, for example.

[0027] Referring to FIGS. 1 and 5, the fastener-receiving sub-assembly **34** may include the extender guide **40**. The extender guide **40** may be structured to be attachable to the connecting element **38** using fasteners or other methods. The extender guide **40** may be structured to rotatably support one or more extenders **43** (described below) usable to rotate the threaded fasteners **71** that secure the shock absorber second end **23b** to the shock tower **26**.

[0028] In one or more arrangements, the extender guide **40** may include a base portion **40b** and one or more extender passages **40p** formed in the base portion **40b**. Each extender passage **40p** may have a central axis XX. The example shown in the drawings includes a pair of extender passages **40p** formed therein. The extender passages **40p** may include sleeve bearings **40r** mounted therein to facilitate rotation of the portions of extenders **43** residing inside the passages **40p**.

[0029] The extender guide **40** may include a bearing surface **40s** structured to abut and support the connecting element **38** when the extender guide **40** is attached to the connecting element. The extender guide **40** may be attached to the connecting element **38** using threaded fasteners or any

other suitable method. The extender guide base portion **40b** may include through holes **40h** enabling attachment of the extender guide **40a** to the connecting element **38** using threaded fasteners. The extender guide base portion **40b** may also include a hole **40y** structured to receive therein the locating pin **38p** of connecting element **38**, to aid in positioning the extender guide **40** with respect to the connecting element **38**. The extender guide **40** may be formed from a relatively rigid material, such as a suitable polymer, for example.

[0030] Referring to FIGS. 1 and 6, the fastener-receiving sub-assembly **34** may include at least one extender assembly **42**. In the examples described herein, the fastener-receiving sub-assembly **34** includes a pair of extender assemblies **42**. Each extender assembly **42** may be structured to be rotatably supported by the extender guide **40** in an associated one of extender passages **40p**.

[0031] Each extender assembly **42** may include an extender **43**. Each extender **43** may be cylindrical and may have a first end **43a**, a second end **43b** opposite the first end, and a body **43c** extending between and connecting the first and second ends **43a**, **43b**. Each extender **43** may have a longitudinal central axis **X3**. In some arrangements, the extender **43** may be a cylindrical rod formed from a suitable material (e.g., aluminum). Each extender **43** and the extender guide **40** may be structured so that the extender is rotatable within, and axially slidable along, an interior of an associated extender passage **40p** of the extender guide **40**.

[0032] Referring to FIGS. 1 and 6, each extender assembly **42** may include a socket **44** defining an interior cavity **44a** having a central axis **X2** and structured for receiving a fastener **71** therein. Each socket **44** may be structured to be attachable to a first end **43a** of an associated extender **43** so as to rotate in correspondence with the extender **43** (i.e., a fastener **71** received in the socket **44** will rotate simultaneously with, and by the same amount as, the socket). Each socket may be structured to be attachable to an end of an associated extender **43** so that a central axis **X2** of its socket cavity **44a** is coaxial (within applicable tolerance limits) with a central axis **X3** of the associated extender.

[0033] To enable the fastener **71** to be rotated by the socket **44** and in correspondence with the socket, the socket interior cavity **44a** may be shaped to conform to a geometry of the fastener **71** to be received in the cavity **44a**. Each cavity **44a** may be structured so that a central axis **XF** of a fastener **71** received in the cavity will be coaxial (within applicable tolerance limits) with the central axis **X2** of the cavity **44a**. In one or more arrangements, the socket **44** and the fastener **71** to be received in the socket may be structured so that the fastener **71** forms a close sliding fit with walls of the fastener receiving cavity **44a**. In some arrangements, the sockets **44** may be magnetized for retaining any fasteners **71** received in the socket cavities **44a**.

[0034] In embodiments described herein, the fastener-receiving sub-assembly **34** may be structured to be rotatable with respect to the cap body portion **32b** so as to enable movement of at least one fastener **71** received in the sub-assembly **34** into an assembly orientation of the fastener **71** when the sub-assembly **34** is attached to the cap flap **32f** and the cap body portion **32b** is attached to a shock absorber **23**. An assembly orientation of the fastener **71** may be an orientation of the fastener with respect to the shock absorber in which fastener **71** need only be driven some distance along a central, longitudinal axis **XF** of the fastener toward

a respective attachment ear **23e** to apply the fastener to the shock absorber, when attaching the shock absorber **23** to the vehicle body. The assembly orientation of a fastener **71** may be achieved when a central axis **XF** of the fastener is coaxial (within applicable tolerance limits) with the central axis **XH** of a through hole **23h** of an associated shock absorber attachment ear **23e**, and the central axis **XF** of the fastener **71** is perpendicular (within applicable tolerance limits) to the plane **P1** of the attachment ears. Referring to FIG. 7C, to aid in achieving and maintaining the fasteners **71** in their respective assembly orientations, the bracing portion **36**, connecting element **38**, and extender guide **40** may be structured so that the central axis **XX** of each extender guide passage extends coaxially (within applicable tolerance limits) with a central axis **XX** of an associated through hole **23h** of the shock absorber attachment ear **23e** when the bracing portion **36**, connecting element **38**, and extender guide **40** are structured and attached to each other as described herein, when the fastener-receiving sub-assembly **34** is attached to the cap flap **32f** and the cap body portion **32b** is attached to a shock absorber **23** as described herein, and when at least the bracing portion second shoulder **36b** is rotated so as to be in physical contact with the shock absorber body **23c**.

[0035] Operation of the shock absorber attachment jig **30** will now be discussed with reference to the drawings.

[0036] Referring to FIGS. 6 and 7A-7C, with the fastener-receiving sub-assembly **34** attached to the cap flap **32f** as shown in FIG. 6, and after the shock absorber **23** has been connected to an associated wheel axle or another portion of the vehicle frame **22** as shown in FIG. 1, the cap body portion **32b** of the jig **30** may be applied to the second end **23b** of the shock absorber as previously described. Because the cap **32** is formed from a relatively flexible material, the cap flap **32f** and the attached fastener-receiving sub-assembly **34** may be rotated away from the shock absorber body **23c** (FIG. 7A) as the cap body portion **32b** is applied to the shock absorber **23**. After the cap body portion **32b** is attached to the shock absorber **23**, a fastener **71** may be inserted into each of socket cavities **44a** (FIG. 7B) for attachment of the shock absorber to the shock tower. The fastener-receiving sub-assembly **34** may then be released so that it drops into a position where the bracing portion first and second shoulders **36a**, **36b** come into physical contact with the shock absorber body **23c**, with the shock absorber flange **23f** received in the bracing portion cavity **36d** (FIG. 7C). Alternatively, the fastener-receiving sub-assembly **34** may be manually rotated toward the shock absorber body **23c** until at least the bracing portion second shoulder **36b** physically contact the shock absorber body **23c**, with the shock absorber flange **23f** engaging extending into the bracing portion cavity **36d** and engaging surface(s) of the bracing portion **36** in an interference fit to maintain the bracing portion second shoulder **36b** in physical contact with the shock absorber body **23c**. This process orients the socket cavities **44a** so that the fasteners **71** received in the socket cavities will be in their respective assembly orientations when the fasteners are positioned in the cavities.

[0037] Referring to FIGS. 1 and 7C, to attach the shock absorber **23** to the shock tower **26**, the vehicle frame **22** and body **25** may be positioned with respect to each other so that the shock absorber attachment ear holes **23h** and the fasteners **71** in the sockets **44** are aligned with associated threaded fastener-receiving holes in the shock tower **26**. if needed, the shock absorber **23** may be pivoted about its first end **23a**

previously attached to the vehicle frame 22, to move the shock absorber second end 23b into a position where the fasteners 71 are in a position ready for the fasteners 71 to be applied to the shock tower 26. An impact driver or other suitable tool (not shown) may then be applied to the second end 43b of each extender 43 to rotate the extender assembly 42 while moving the extender assembly in a direction DI toward the shock tower 26, until the fasteners 71 extend through the attachment ear holes 23h and into the shock tower 26. The shock absorber attachment jig 30 may then be removed from the shock absorber.

[0038] Embodiments of the shock absorber attachment jig described herein may enable the threaded fasteners used to attach the shock absorber to the vehicle body to be positioned and secured in an orientation where they may be conveniently aligned for application to the shock absorber and shock tower by simple rotation of an end of the shock absorber.

[0039] In the above detailed description, reference is made to the accompanying figures, which form a part hereof. In the figures, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, figures, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

[0040] The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e. open language). The phrase “at least one of . . . and . . .” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B and C” includes A only, B only, C only, or any combination thereof (e.g. AB, AC, BC or ABC).

[0041] Aspects herein can be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope of the invention.

What is claimed is:

1. A shock absorber attachment jig comprising:

a cap including a body portion and a flap extending from the body portion; and

a fastener-receiving sub-assembly structured to be attached to the flap,

wherein the flap is structured to be rotatable with respect to the body portion so as to enable rotation of the sub-assembly with respect to the cap body portion when the sub-assembly is attached to the flap.

2. The shock absorber attachment jig of claim 1, wherein the sub-assembly is structured to be rotatable with respect to the cap body portion so as to enable movement of a fastener received in the sub-assembly to an assembly orientation of the fastener when the sub-assembly is attached to the flap and the cap body portion is attached to a shock absorber.

3. The shock absorber attachment jig of claim 1, wherein the sub-assembly includes a bracing portion defining a cavity structured to receive therein a flange of a shock absorber.

4. The shock absorber attachment jig of claim 3, wherein the flap is structured to be rotatable to position the flange inside the cavity when the sub-assembly is attached to the flap and the cap body portion is attached to the shock absorber.

5. The shock absorber attachment jig of claim 3, wherein the bracing portion includes a first shoulder, a second shoulder opposite the first shoulder, and a wall connecting the first and second shoulders, the first shoulder, the second shoulder and the wall defining the cavity.

6. The shock absorber attachment jig of claim 2, wherein the sub-assembly includes a bracing portion structured to be rotatable into physical contact with the shock absorber, and wherein the jig is structured so that a fastener received in the sub-assembly resides in the assembly orientation when the sub-assembly is attached to the flap, the cap body portion is attached to a shock absorber, and the bracing portion is in physical contact with the shock absorber.

7. The shock absorber attachment jig of claim 2, wherein the sub-assembly includes:

an extender assembly structured to support the fastener; and

an extender guide structured to support the extender assembly so that the extender assembly is rotatable and axially movable within an associated extender cavity formed in the extender guide.

* * * * *